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Project: Power Bill: Winter Residential Rates

Company: Dominion Energy South Carolina

https://cdn-dominionenergy-prd-001.azureedge.net/-/media/pdfs/south-carolina/rates-and tariffs/rate1.pdf?la=en&rev=b48025a85dc34e49b550a2e625ea0a12&hash=5410E8F66F904B6F294F3A7AEB7AD880

Objectives:

(1.) Calculate the power bill of the residential service for the residents of South Carolina within each range of specific power usage manually using the Arithmetic method for the Winter.

(2.) Write a piecewise function for the Winter rates: October-May rates.

(3.) Recalculate the same power bill of residential service for the residents of South Carolina within each range of specific power usage algebraically using the Piecewise Function method for the Winter.

Information:

<u>Winter</u>

(Billing Months: October - May)

Basic Facilities Charge: $9.00

Plus Facilities Charge:

First 800 kWh @ $0.13367 per kWh

Excess over 800 kWh @ $0.12921 per kWh

Arithmetic Method:

I am going to calculate the costs for these power consumptions:

(1.) 0 kWh
(2.) 700 kWh
(3.) 1200 kWh

(1.) For 0 kWh:

$$Cost = \$9.00$$

(2.) For 700 kWh:

$$Cost = 9.00 + 0.13367(700)$$

$$Cost = 9 + 93.569$$

$Cost = 102.569$

$Cost \cong \$102.57$

(3.) For 1200 kWh:

$Cost = 9.00 + 0.13367(800) + 0.12921(1200 - 800)$

$Cost = 9 + 106.936 + 0.12921(400)$

$Cost = 9 + 106.936 + 51.684$

$Cost = \$167.62$

Piecewise Function:

Define Variables:

Let p = power consumed in kWh

Let C = cost of power consumed per kWh in $

Minimum Charge = $9.00

First Piece:

Cost for $0 \leq p \leq 800$ is 0.13367 per kWh

$C(p) = 9 + 0.13367p$

$C(p) = 0.13367p + 9$

Second Piece:

Minimum Charge = \$9.00

Cost for $p > 800$ is 0.12921 per kWh

Cost for 800 kWh = 0.13367(800) = 106.936

$C(p) = 9 + 106.936 + 0.12921(p - 800)$

$C(p) = 115.936 + 0.12921p - 103.368$

$C(p) = 0.12921p + 12.568$

$$C(p) = \begin{cases} 0.13367p + 9; & 0 \leq p \leq 800 \\ 0.12921p + 12.568; & p > 800 \end{cases}$$

Piecewise Function Method:

(1.) For 0 kWh: First Piece

$C(p) = 0.13367p + 9$

$C(0) = 0.13367(0) + 9$

$C(0) = \$9.00$

(2.) For 700 kWh: First Piece

$C(p) = 0.13367p + 9$

$C(700) = 0.13367(700) + 9$

$C(700) = 102.569$

$C(700) \cong \$102.57$

(3.) For 1200 kWh: Second Piece

$C(p) = 0.12921p + 12.568$

$C(1200) = 0.12921(1200) + 12.568$

$C(1200) = \$167.62$

Works Cited (MLA)

DOMINION ENERGY SOUTH CAROLINA, INC. ELECTRICITY RATE 1 RESIDENTIAL SERVICE GOOD CENTS RATE (Page 1 of 2). cdn-dominionenergy-prd-001.azureedge.net/-/media/pdfs/south-carolina/rates-and-tariffs/rate1.pdf?la=en&rev=b48025a85dc34e49b550a2e625ea0a12&hash=5410E8F66F904B6F294F3A7AEB7AD880. Accessed 24 Apr. 2024.